Solving Systems using Matrices



A **Matrix** is a rectangular array of numbers arranged in rows and columns.

They can be used to represent and solve linear systems.

To Convert a linear system to matrix notation:

1) Each row will correspond to an EQUATION.

2) Each column will correspond to a VARIABLE / CONSTANT.

A linear system of equations can be written in a matrix form as follows:

$$\begin{aligned} 2x-6y+4z&=11\\ x-3y+4z&=-7\\ 8x+18y-2z&=-1 \end{aligned}$$

$$\begin{bmatrix} 2 & -6 & 4\\ 1 & -3 & 4\\ 8 & 18 & -2 \end{bmatrix}\begin{bmatrix} 11\\ -7\\ -1 \end{bmatrix}$$

(x, y, z; column of constants; "element"; coefficient matrix)

OR

$$\left[\begin{array}{ccc|c} 2 & -6 & 4 & 11\\ 1 & -3 & 4 & -7\\ 8 & 18 & -2 & -1 \end{array}\right]$$

("=" ; "AUGMENTED MATRIX")

Example 1: Write each of the following linear systems in augmented matrix form.

a) $\begin{aligned} 5x-y+3z&=2\\ x-4y+z&=-5\\ 2x+5y-2z&=-1 \end{aligned}$

$$\left[\begin{array}{ccc|c} 5 & -1 & 3 & 2\\ 1 & -4 & 1 & -5\\ 2 & 5 & -2 & -1 \end{array}\right]$$

(x, y, z, constant)

b) $x+3z-1=0$ ← not lined up! $\Rightarrow x+0y+3z=1$

$x-4y+5z=0$

$$\left[\begin{array}{ccc|c} 1 & 0 & 3 & 1\\ 1 & -4 & 5 & 0 \end{array}\right]$$

Example 2: Solve the following linear system using matrices.

$$\begin{aligned} 2x-5y+z&=3\\ x-3y-2z&=-9\\ 3x-6y-2z&=-8 \end{aligned}$$

The process of row-reducing is the same as performing operations to create equivalent systems.

$$\left[\begin{array}{ccc|c} 2 & -5 & 1 & 3\\ 1 & -3 & -2 & -9\\ 3 & -6 & -2 & -8 \end{array}\right]$$

It's easier if leading coefficient is 1!!

$$\xrightarrow{\begin{matrix} R2\\ R1\\ \end{matrix}}\left[\begin{array}{ccc|c} 1 & -3 & -2 & -9\\ 2 & -5 & 1 & 3\\ 3 & -6 & -2 & -8 \end{array}\right]\xrightarrow{\begin{matrix} 2R1-R2\\ 3R1-R3 \end{matrix}}\left[\begin{array}{ccc|c} 1 & -3 & -2 & -9\\ 0 & -1 & -5 & -21\\ 0 & -3 & -4 & -19 \end{array}\right]$$

eliminate x!

$$\xrightarrow{\begin{matrix} \\ -R2\\ -R3 \end{matrix}}\left[\begin{array}{ccc|c} 1 & -3 & -2 & -9\\ 0 & 1 & 5 & 21\\ 0 & 3 & 4 & 19 \end{array}\right]\xrightarrow{3R2-R3}\left[\begin{array}{ccc|c} 1 & -3 & -2 & -9\\ 0 & 1 & 5 & 21\\ 0 & 0 & 11 & 44 \end{array}\right]$$

eliminate y!

From R3: $11z=44$

$z=4$

Sub in R2: $y+5z=21$

$y+5(4)=21$

$y=1$

Sub in R1: $x-3y-2z=-9$

$x-3(1)-2(4)=-9$

$x=2$

$\therefore$ P.O.I. $(2,1,4)$

Row-Echelon Form:

A matrix form in which each subsequent row has its first non-zero entry further to the right than in the preceding row.

$$\begin{bmatrix}1&2&3\\4&5&6\\6&7&8\end{bmatrix} \xrightarrow{\text{"Row-Reducing" operations}} \begin{bmatrix}1&2&3\\0&1&2\\0&0&0\end{bmatrix}$$

Gaussian Elimination.

Some of the operations include:

1. INTERCHANGE two rows "switch rows"
2. Replace a row with a scalar multiples of itself "multiply/divide" by a constant.
3. Replace a row with a linear combination of any two others
   e.g. $2R1 - R2$.

Examples:

$$\begin{bmatrix}8&2&5&-3\\0&2&9&1\\0&0&-4&2\\0&0&0&7\end{bmatrix}$$

triangle of zeros.

Row-Echelon Form

Really easy to see the solution.

$$\left[\begin{array}{ccc|c}1&0&0&2\\0&1&0&-1\\0&0&1&3\end{array}\right] \Rightarrow x=2,\ y=-1,\ z=3.$$

Reduced Row-Echelon Form.
- "diagonals of ones" ONLY
- every other elements are zeroes.

Example 3: Solve the following linear system using matrices.

$$3x+5y-z=6$$
$$2x+y+4z=4$$
$$6x+10y-2z=12$$

$$\left[\begin{array}{ccc|c}3&5&-1&6\\2&1&4&4\\6&10&-2&12\end{array}\right] \xrightarrow[2R1-R3]{2R1-3R2} \left[\begin{array}{ccc|c}3&5&-1&6\\0&7&-14&0\\0&0&0&0\end{array}\right]$$

→ entire row of zero !!!

∴ Infinite solution.

Let $z=t$, $t\in\mathbb{R}$.

Sub in R2: $7y-14z=0$

$7y-14t=0$

$7y=14t$

$y=2t$

sub in R1: $3x+5y-z=6$

$3x+5(2t)-(t)=6$

$3x+9t=6$

$3x=6-9t$

$x=2-3t$

$\therefore (x,y,z)=(2-3t,\ 2t,\ t)$

$=(2,0,0)+t(-3,2,1),\ t\in\mathbb{R}$

← intersection is a LINE !!